

SOC Internship – Day 22 Notes

SIEM Advanced Use Cases

Objective

Learn how SOC teams build advanced SIEM detection rules for real-world attack detection.

1. What is a SIEM Use Case?

A **SIEM Use Case** is a detection logic or rule created inside a SIEM platform to identify suspicious, malicious, or abnormal activities using collected logs and events.

Simple Definition

It helps SOC analysts automatically detect cyber threats from system logs.

Example

- Multiple failed logins
- Suspicious PowerShell commands
- Unauthorized privilege escalation

2. Importance of SIEM Use Cases

SIEM use cases are important because they help SOC teams:

- Detect advanced cyber threats
- Reduce false positives
- Improve security monitoring
- Automate threat detection and alerting

Benefits

- Faster incident response
- Better visibility
- Reduced manual monitoring

- Improved threat hunting

3. Common Advanced SIEM Use Cases

a) Brute Force Login Detection

Detects repeated failed login attempts.

Indicators

- Multiple failed logins
- Same source IP
- Short time interval

Risk

Attackers trying to guess passwords.

b) Impossible Travel Detection

Detects logins from two geographically distant locations within an impossible time.

Example

- Login from India
- 10 minutes later login from USA

Risk

Compromised user account.

c) Privilege Escalation Alerts

Detects when a normal user suddenly receives admin rights.

Risk

Potential insider threat or compromised account.

d) Suspicious PowerShell Execution

Detects malicious or encoded PowerShell commands.

Indicators

- Encoded commands
- Download scripts
- Remote execution

Risk

Malware execution or attacker activity.

e) Data Exfiltration Monitoring

Detects large or unusual data transfers.

Indicators

- Large downloads
- External transfers
- Sensitive file access

Risk

Data theft or insider attack.

4. Detection Logic Example

Example SIEM rule:

IF failed_logins > 10
AND source_ip is external
THEN generate brute force alert

Explanation

- More than 10 failed logins
- From an external IP address

- SIEM generates an alert automatically

5. Correlation Example

Events

1. User login
2. Admin privilege granted
3. Sensitive files downloaded

Result

Potential insider threat detected.

What is Correlation?

Correlation means connecting multiple events together to identify a bigger security incident.

Why Important?

Single events may look normal, but combined events can reveal attacks.

6. SOC Workflow in SIEM Monitoring

L1 Analyst

- Monitor alerts
- Identify suspicious events
- Escalate incidents

L2 Analyst

- Investigate suspicious activity
- Analyze logs and evidence
- Confirm attacks

L3 Analyst

- Improve SIEM rules
- Fine-tune detections
- Create advanced use cases

7. SIEM Challenges

High Log Volume

Large amount of logs generated daily.

False Positives

Benign activity incorrectly flagged as malicious.

Missing Logs

Some systems may not send logs properly.

Complex Attack Patterns

Advanced attacks are difficult to detect.

Class Activity Answer

Scenario

- 10 failed logins
- 1 successful login
- Large data download

Which SIEM Use Case Should Trigger?

Answer:

Potential Account Compromise / Brute Force + Data Exfiltration Alert

Reason

- Multiple failed logins indicate brute force attack
- Successful login may mean attacker gained access
- Large data download suggests possible data theft

Assignment Answers

1. Define SIEM Use Case

A SIEM use case is a rule or detection logic used to identify suspicious or malicious activities using logs and security events.

2. Create a Detection Rule

```
IF failed_login_attempts > 10  
AND login_success = TRUE  
AND large_file_download = TRUE  
THEN generate account compromise alert
```

3. Explain Correlation Logic

Correlation logic combines multiple related events to detect advanced attacks.
For example:

- Failed logins
- Successful login
- Large data download

Together indicate a possible compromised account and data exfiltration attempt.